

GAUTAM KAMATH

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RESEARCH INTERESTS

Reliable and trustworthy algorithms, statistics, and machine learning, particularly privacy and robustness.

PROFESSIONAL APPOINTMENTS

New York University

Computer Science, Courant Institute of Mathematical Sciences
Assistant Professor
September 2026 - Present

University of Waterloo

David R. Cheriton School of Computer Science
Assistant Professor
July 2019 - Present (on leave from September 2026)

Vector Institute

Faculty Member, Canada CIFAR AI Chair
March 2023 - Present
Faculty Affiliate
December 2020 - March 2023

Simons Institute for the Theory of Computing

Microsoft Research Fellow
August 2018 - May 2019

EDUCATION

Massachusetts Institute of Technology

Ph.D., September 2018
S.M., September 2014
Electrical Engineering and Computer Science

Cornell University

B.S., summa cum laude, May 2012
Computer Science, Electrical and Computer Engineering

SELECTED HONOURS AND AWARDS

Gödel Prize	July 2026
<i>For “Robust Estimators in High Dimensions without the Computational Intractability”</i>	
<i>For outstanding papers in the area of theoretical computer science</i>	
Presburger Award	July 2026
<i>To a young scientist for outstanding contributions in theoretical computer science</i>	
Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies,	July 2026
Runner Up	
<i>For “Differentially Private Fine-tuning of Language Models”</i>	
<i>Awarded annually to 1-2 papers for outstanding contributions to privacy enhancing technologies</i>	
Ontario Early Researcher Award	April 2025
<i>To new researchers working at publicly funded Ontario research institutions, based on research excellence</i>	
Best Paper Award, International Conference on Machine Learning (ICML 2024)	July 2024
<i>For “Position: Considerations for Differentially Private Learning with Large-Scale Public Pretraining”</i>	
<i>Awarded to 10 best papers, out of 2610 accepted and 9473 submitted papers</i>	
Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies	July 2024
<i>For “The Discrete Gaussian for Differential Privacy”</i>	
<i>Awarded annually to 1-2 papers for outstanding contributions to privacy enhancing technologies</i>	
University of Waterloo Faculty of Math Golden Jubilee Research Excellence Award	July 2023

<i>For outstanding research contributions, early-career recipient</i>	March 2023
Canada CIFAR AI Chair	
<i>To recruit and retain world leading AI researchers to Canada</i>	
NSERC Discovery Accelerator Supplement	April 2020
<i>To accelerate an established, superior research program</i>	

SELECTED PROFESSIONAL ACTIVITIES

IEEE Conference on Secure and Trustworthy Machine Learning, Steering Committee	July 2026 - Present
Symposium on Foundations of Responsible Computing, Executive Committee	November 2025 - Present
International Conference on Machine Learning, Board Member	July 2025 - Present
Association for Computational Learning, Director	July 2024 - Present
36th International Conference on Algorithmic Learning Theory (ALT 2025), PC Co-Chair	June 2024 - February 2025
Association for Algorithmic Learning Theory, Steering Committee	June 2024 - February 2029
Learning Theory Alliance, Executive Committee	November 2023 - Present
Transactions on Machine Learning Research (TMLR), Editor in Chief	July 2023 - December 2026
52nd Annual ACM Symposium on Theory of Computing (STOC 2020), General Chair	March 2020 - June 2020

PUBLICATIONS

Metrics: **7307** citations, h-index **41** (according to Google Scholar, August 22, 2026)

Primary publication venues: NeurIPS, ICML, COLT, STOC, FOCS, SODA

In computer science, conference proceedings are the primary venue for publishing complete research works.

Most authorships are in alphabetical order. Papers with contribution-order authorship are indicated, and equal contributions are marked with * or ^. Generally, these will put the students as first-author, with equal contribution amongst the senior authors. ® is used for randomized author order.

Demystifying Foreground-Background Memorization in Diffusion Models

Jimmy Z. Di*, Yiwei Lu*, Yaoliang Yu^, Gautam Kamath^, Adam Dziedzic^, Franziska Boenisch^ (Contribution order)

Proceedings of the Fortieth AAAI Conference on Artificial Intelligence (AAAI 2026)

Rethinking Benchmarks for Differentially Private Image Classification

Sabrina Mokhtari*, Sara Kodeiri*, Shubhankar Mohapatra*, Florian Tramèr^, Gautam Kamath^ (Contribution order)

IEEE Data Engineering Bulletin, 49(4), 2025

A Bias-Accuracy-Privacy Trilemma for Statistical Estimation

Gautam Kamath, Argyris Mouzakis, Matthew Regehr, Vikrant Singhal, Thomas Steinke, Jonathan Ullman

Journal of the American Statistical Association, 120(552), 2025

Query-Efficient Locally Private Hypothesis Selection via the Scheffe Graph

Gautam Kamath, Alireza F. Pour, Matthew Regehr, David Woodruff

Advances in Neural Information Processing Systems 38 (NeurIPS 2025)

BridgePure: Limited Protection Leakage Can Break Black-Box Data Protection

Yihan Wang*, Yiwei Lu*, Xiao-Shan Gao^, Gautam Kamath^, Yaoliang Yu^ (Contribution order)

Advances in Neural Information Processing Systems 38 (NeurIPS 2025)

Optimal Differentially Private Sampling of Unbounded Gaussians

Valentio Iverson, Gautam Kamath, Argyris Mouzakis

Proceedings of the 38th Annual Conference on Learning Theory (COLT 2025)

On the Learnability of Distribution Classes with Adaptive Adversaries

Tosca Lechner, Alex Bie*, Gautam Kamath* (Contribution order)

Proceedings of the 42nd International Conference on Machine Learning (ICML 2025)

Machine Unlearning Fails to Remove Data Poisoning Attacks

Martin Pawelczyk*, Jimmy Z. Di*, Yiwei Lu, Gautam Kamath^, Ayush Sekhari^, Seth Neel^ (Contribution

order)

Proceedings of the 13th International Conference on Learning Representations (ICLR 2025)

Position: Membership Inference Attacks Cannot Prove that a Model Was Trained On Your Data

Jie Zhang, Debeshee Das, Gautam Kamath, Florian Tramèr (Contribution order)

Proceedings of the 2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML 2025)

Avoiding Pitfalls for Privacy Accounting of Subsampled Mechanisms under Composition

Christian Janos Lebeda*, Matthew Regehr*, Gautam Kamath[^], Thomas Steinke[^] (Contribution order)

Proceedings of the 2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML 2025)

Choosing Public Datasets for Private Machine Learning via Gradient Subspace Distance

Xin Gu, Gautam Kamath*, Zhiwei Steven Wu* (Contribution order)

Proceedings of the 2025 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML 2025)

Private Mean Estimation with Person-Level Differential Privacy

Sushant Agarwal, Gautam Kamath, Mahbod Majid, Argyris Mouzakis, Rose Silver, Jonathan Ullman

Proceedings of the 36th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2025)

Unbiased Statistical Estimation and Valid Confidence Intervals Under Differential Privacy

Christian Covington, Xi He*, James Honaker*, Gautam Kamath* (Contribution order)

Statistica Sinica, 35(Online Special Issue I), 2025

The Broader Landscape of Robustness in Algorithmic Statistics

Gautam Kamath

IEEE BITS the Information Theory Magazine, 4(3), 2024

Differentially Private Post-Processing for Fair Regression

Ruicheng Xian, Qiaobo Li, Gautam Kamath, Han Zhao (Contribution order)

Proceedings of the 41st International Conference on Machine Learning (ICML 2024)

Disguised Copyright Infringement of Latent Diffusion Models

Yiwei Lu*, Matthew Y.R. Yang*, Zuoqiu Liu*, Gautam Kamath[^], Yaoliang Yu[^] (Contribution order)

Proceedings of the 41st International Conference on Machine Learning (ICML 2024)

Position: Considerations for Differentially Private Learning with Large-Scale Public Pretraining

Florian Tramèr*, Gautam Kamath*, Nicholas Carlini* (Reverse alphabetical order)

Proceedings of the 41st International Conference on Machine Learning (ICML 2024)

ICML 2024 Best Paper

Oral Presentation

Indiscriminate Data Poisoning Attacks on Pre-trained Feature Extractors

Yiwei Lu, Matthew Y.R. Yang, Gautam Kamath*, Yaoliang Yu* (Contribution order)

Proceedings of the 2024 IEEE Conference on Secure and Trustworthy Machine Learning (SaTML 2024)

Advancing Differential Privacy: Where We Are Now and Future Directions for Real-World Deployment

Rachel Cummings, Damien Desfontaines, David Evans, Roxana Geambasu, Yangsibo Huang, Matthew Jagielski, Peter Kairouz, Gautam Kamath, Sewoong Oh, Olga Ohrimenko, Nicolas Papernot, Ryan Rogers, Milan Shen, Shuang Song, Weijie Su, Andreas Terzis, Abhradeep Thakurta, Sergei Vassilvitskii, Yu-Xiang Wang, Li Xiong, Sergey Yekhanin, Da Yu, Huan Yu Zhang, Wanrong Zhang

Harvard Data Science Review, 6(1), 2024

Not All Learnable Distribution Classes are Privately Learnable

Mark Bun, Gautam Kamath, Argyris Mouzakis, Vikrant Singhal

Private Distribution Learning with Public Data: The View from Sample Compression

Shai Ben-David, Alex Bie, Clément L. Canonne, Gautam Kamath, Vikrant Singhal

Advances in Neural Information Processing Systems 36 (NeurIPS 2023)

Spotlight Presentation

Distribution Learnability and Robustness

Shai Ben-David, Alex Bie, Gautam Kamath, Tosca Lechner

Advances in Neural Information Processing Systems 36 (NeurIPS 2023)

Hidden Poison: Machine Unlearning Enables Camouflaged Poisoning Attacks

Jimmy Z. Di, Jack Douglas, Jayadev Acharya*, Gautam Kamath*, Ayush Sekhari* (Contribution order)

Advances in Neural Information Processing Systems 36 (NeurIPS 2023)

Private GANs, Revisited

Alex Bie, Gautam Kamath*, Guojun Zhang* (Contribution order)

Transactions on Machine Learning Research (TMLR), 2023

Survey Certification

Individual Privacy Accounting for Differentially Private Stochastic Gradient Descent

Da Yu, Gautam Kamath*, Janardhan Kulkarni*, Tie-Yan Liu*, Jian Yin*, Huishuai Zhang* (Contribution order)

Transactions on Machine Learning Research (TMLR), 2023

Exploring the Limits of Model-Targeted Indiscriminate Data Poisoning Attacks

Yiwei Lu, Gautam Kamath*, Yaoliang Yu*. (Contribution order)

Proceedings of the 40th International Conference on Machine Learning (ICML 2023)

Robustness Implies Privacy in Statistical Estimation

Samuel B. Hopkins, Gautam Kamath, Mahbod Majid, Shyam Narayanan

Proceedings of the 55th ACM Symposium on Theory of Computing (STOC 2023)

Indiscriminate Data Poisoning Attacks on Neural Networks

Yiwei Lu, Gautam Kamath*, Yaoliang Yu* (Contribution order)

Transactions on Machine Learning Research (TMLR), 2022

New Lower Bounds for Private Estimation and a Generalized Fingerprinting Lemma

Gautam Kamath, Argyris Mouzakis, Vikrant Singhal

Advances in Neural Information Processing Systems 35 (NeurIPS 2022)

Private Estimation with Public Data

Alex Bie, Gautam Kamath, Vikrant Singhal

Advances in Neural Information Processing Systems 35 (NeurIPS 2022)

Improved Rates for Differentially Private Stochastic Convex Optimization with Heavy-Tailed Data

Gautam Kamath, Xingtu Liu, Huanyu Zhang

Proceedings of the 39th International Conference on Machine Learning (ICML 2022)

Long Talk

Robust Estimation for Random Graphs

Jayadev Acharya, Ayush Jain, Gautam Kamath, Ananda Theertha Suresh, Huanyu Zhang

Proceedings of the 35th Annual Conference on Learning Theory (COLT 2022)

A Private and Computationally-Efficient Estimator for Unbounded Gaussians

Gautam Kamath, Argyris Mouzakis, Vikrant Singhal, Thomas Steinke, Jonathan Ullman

Proceedings of the 35th Annual Conference on Learning Theory (COLT 2022)

The Price of Tolerance in Distribution Testing

Clément L. Canonne, Ayush Jain, Gautam Kamath, Jerry Li

Proceedings of the 35th Annual Conference on Learning Theory (COLT 2022)

Calibration with Privacy in Peer Review

Wenxin Ding, Gautam Kamath @ Weina Wang @ Nihar B. Shah (Contribution order, with randomization)

Proceedings of the 2022 IEEE International Symposium on Information Theory (ISIT 2022)

Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism

Samuel B. Hopkins, Gautam Kamath, Mahbod Majid

Proceedings of the 54th ACM Symposium on Theory of Computing (STOC 2022)

Presented at the 3rd Symposium on Foundations of Responsible Computing (FORC 2022, non-archival track)

Differentially Private Fine-tuning of Language Models

Da Yu, Saurabh Naik, Arturs Backurs*, Sivakanth Gopi*, Huseyin A. Inan*, Gautam Kamath*, Janardhan Kulkarni*, Yin Tat Lee*, Andre Manoel*, Lukas Wutschitz*, Sergey Yekhanin*, Huishuai Zhang* (Contribution order)

Journal of Privacy and Confidentiality, 14(2), 2024

Proceedings of the 10th International Conference on Learning Representations (ICLR 2022)

2026 Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies, Runner-Up

The Role of Adaptive Optimizers for Honest Private Hyperparameter Selection

Shubhankar Mohapatra*, Sajin Sasy*, Xi He[^], Gautam Kamath[^], Om Thakkar[^] (Contribution order)

Proceedings of the Thirty-Sixth AAAI Conference on Artificial Intelligence (AAAI 2022)

Oral Presentation

Robustness Meets Algorithms

Ilias Diakonikolas, Gautam Kamath, Daniel M. Kane, Jerry Li, Ankur Moitra, Alistair Stewart

Communications of the ACM, 64(5), 2021

Invited Research Highlight

Remember What You Want to Forget: Algorithms for Machine Unlearning

Ayush Sekhari, Jayadev Acharya*, Gautam Kamath*, Ananda Theertha Suresh* (Contribution order)

Advances in Neural Information Processing Systems 34 (NeurIPS 2021)

Enabling Fast Differentially Private SGD via Just-in-Time Compilation and Vectorization

Pranav Subramani*, Nicholas Vadivelu*, Gautam Kamath (Contribution order)

Advances in Neural Information Processing Systems 34 (NeurIPS 2021)

PAPRIKA: Private Online False Discovery Rate Control

Wanrong Zhang, Gautam Kamath*, Rachel Cummings* (Contribution order)

Proceedings of the 38th International Conference on Machine Learning (ICML 2021)

Presented at the 2nd Symposium on Foundations of Responsible Computing (FORC 2021, non-archival track)

On the Sample Complexity of Privately Learning Unbounded High-Dimensional Gaussians

Ishaq Aden-Ali, Hassan Ashtiani, Gautam Kamath

Proceedings of the 32nd International Conference on Algorithmic Learning Theory (ALT 2021)

Random Restrictions of High-Dimensional Distributions and Uniformity Testing with Subcube Conditioning

Clément L. Canonne, Xi Chen, Gautam Kamath, Amit Levi, Erik Waingarten

Proceedings of the 32nd Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2021)

CoinPress: Practical Private Mean and Covariance Estimation

Sourav Biswas, Yihe Dong, Gautam Kamath, Jonathan Ullman

Advances in Neural Information Processing Systems 33 (NeurIPS 2020)

The Discrete Gaussian for Differential Privacy

Clément L. Canonne, Gautam Kamath, Thomas Steinke

Journal of Privacy and Confidentiality, 12(1), 2022

Advances in Neural Information Processing Systems 33 (NeurIPS 2020)

**2024 Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies
Deployed in the 2020 US Census**

Private Identity Testing for High-Dimensional Distributions

Clément L. Canonne, Gautam Kamath, Audra McMillan, Jonathan Ullman, Lydia Zakynthinou

Advances in Neural Information Processing Systems 33 (NeurIPS 2020)

Spotlight Presentation

Privately Learning Markov Random Fields

Huanyu Zhang, Gautam Kamath*, Janardhan Kulkarni*, Zhiwei Steven Wu* (Contribution order)

Proceedings of the 37th International Conference on Machine Learning (ICML 2020)

Private Mean Estimation of Heavy-Tailed Distributions

Gautam Kamath, Vikrant Singhal, Jonathan Ullman

Proceedings of the 33rd Annual Conference on Learning Theory (COLT 2020)

Locally Private Hypothesis Selection

Sivakanth Gopi, Gautam Kamath, Janardhan Kulkarni, Aleksandar Nikolov, Zhiwei Steven Wu, Huanyu Zhang

Proceedings of the 33rd Annual Conference on Learning Theory (COLT 2020)

Differentially Private Algorithms for Learning Mixtures of Separated Gaussians

Gautam Kamath, Or Sheffet, Vikrant Singhal, Jonathan Ullman

Advances in Neural Information Processing Systems 32 (NeurIPS 2019)

Private Hypothesis Selection

Mark Bun, Gautam Kamath, Thomas Steinke, Zhiwei Steven Wu

IEEE Transactions on Information Theory, 67(3), 2021

Advances in Neural Information Processing Systems 32 (NeurIPS 2019)

Sever: A Robust Meta-Algorithm for Stochastic Optimization

Ilias Diakonikolas, Gautam Kamath, Daniel M. Kane, Jerry Li, Jacob Steinhardt, Alistair Stewart

Proceedings of the 36th International Conference on Machine Learning (ICML 2019)

Privately Learning High-Dimensional Distributions

Gautam Kamath, Jerry Li, Vikrant Singhal, Jonathan Ullman

Proceedings of the 32nd Annual Conference on Learning Theory (COLT 2019)

The Structure of Optimal Private Tests for Simple Hypotheses

Clément Canonne, Gautam Kamath, Audra McMillan, Adam Smith, Jonathan Ullman

Proceedings of the 51st ACM Symposium on Theory of Computing (STOC 2019)

Anaconda: A Non-Adaptive Conditional Sampling Algorithm for Distribution Testing

Gautam Kamath, Christos Tzamos

Proceedings of the 30th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2019)

INSPECTRE: Privately Estimating the Unseen

Jayadev Acharya, Gautam Kamath, Ziteng Sun, Huanyu Zhang

Journal of Privacy and Confidentiality, 10(2), 2020

Proceedings of the 35th International Conference on Machine Learning (ICML 2018)

Actively Avoiding Nonsense in Generative Models

Steve Hanneke, Adam Kalai, Gautam Kamath, Christos Tzamos

Proceedings of the 31st Annual Conference on Learning Theory (COLT 2018)

Which Distribution Distances are Sublinearly Testable?

Constantinos Daskalakis, Gautam Kamath, John Wright

Proceedings of the 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2018)

Testing Ising Models

Constantinos Daskalakis, Nishanth Dikkala, Gautam Kamath

IEEE Transactions on Information Theory, 65(11), 2019

Proceedings of the 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2018)

Robustly Learning a Gaussian: Getting Optimal Error, Efficiently

Ilias Diakonikolas, Gautam Kamath, Daniel M. Kane, Jerry Li, Ankur Moitra, Alistair Stewart

Proceedings of the 29th Annual ACM-SIAM Symposium on Discrete Algorithms (SODA 2018)

Concentration of Multilinear Functions of the Ising Model with Applications to Network Data

Constantinos Daskalakis, Nishanth Dikkala, Gautam Kamath

Advances in Neural Information Processing Systems 30 (NIPS 2017)

Being Robust (in High Dimensions) Can Be Practical

Ilias Diakonikolas, Gautam Kamath, Daniel M. Kane, Jerry Li, Ankur Moitra, Alistair Stewart

Proceedings of the 34th International Conference on Machine Learning (ICML 2017)

Priv'IT: Private and Sample Efficient Identity Testing

Bryan Cai, Constantinos Daskalakis, Gautam Kamath

Proceedings of the 34th International Conference on Machine Learning (ICML 2017)

Robust Estimators in High Dimensions without the Computational Intractability

Ilias Diakonikolas, Gautam Kamath, Daniel M. Kane, Jerry Li, Ankur Moitra, Alistair Stewart

Invited to SIAM Journal on Computing Special Issue for FOCS 2016, 48(2), 2019 (SICOMP)

Proceedings of the 57th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2016)

Invited to Highlights of Algorithms 2017 (HALG 2017)

Invited to Communications of the ACM, Research Highlights (CACM)

2026 Gödel Prize

A Size-Free CLT for Poisson Multinomials and its Applications

Constantinos Daskalakis, Anindya De, Gautam Kamath, Christos Tzamos

Proceedings of the 48th ACM Symposium on Theory of Computing (STOC 2016)

Optimal Testing for Properties of Distributions

Jayadev Acharya, Constantinos Daskalakis, Gautam Kamath

Advances in Neural Information Processing Systems 28 (NIPS 2015)

Spotlight Presentation

On the Structure, Covering, and Learning of Poisson Multinomial Distributions

Constantinos Daskalakis, Gautam Kamath, Christos Tzamos

Proceedings of the 56th Annual IEEE Symposium on Foundations of Computer Science (FOCS 2015)

A Chasm Between Identity and Equivalence Testing with Conditional Queries

Jayadev Acharya, Clément Canonne, Gautam Kamath

Theory of Computing, 14(19), 2018

Proceedings of the 19th International Workshop on Randomization and Computation (RANDOM 2015)

Adaptive Estimation in Weighted Group Testing

Jayadev Acharya, Clément Canonne, Gautam Kamath

Proceedings of the 2015 IEEE International Symposium on Information Theory (ISIT 2015)

Faster and Sample Near-Optimal Algorithms for Proper Learning Mixtures of Gaussians

Constantinos Daskalakis, Gautam Kamath

Proceedings of the 27th Annual Conference on Learning Theory (COLT 2014)

An Analysis of One-Dimensional Schelling Segregation

Christina Brandt, Nicole Immorlica, Gautam Kamath, Robert Kleinberg

Proceedings of the 44th ACM Symposium on Theory of Computing (STOC 2012)

PREPRINTS AND OTHER WRITINGS

Efficient and Optimal Learning of Discrete Distributions with Person-Level Privacy

Gautam Kamath, Mahbod Majid, Ankur Moitra, Argyris Mouzakis, Jonathan Ullman

Manuscript

Near-Optimal Pure Machine Unlearning for Smooth Strongly Convex Losses

Matthew Regehr, Gautam Kamath, Andrew Lowy (Contribution order)

Manuscript

Sequential Data Poisoning in LLM Post-Training

Jack Sanderson, Yihan Wang, Xiaoqian Lu, Gautam Kamath*, Yiwei Lu* (Contribution order)

Manuscript

A Private Empirical Defense Against Privacy Audits

Saloni Modi, Srivi Balaji, Yusong Zhu, Gautam Kamath, Kevin Tian (Contribution order)

Manuscript

ContinuousBench: Can Differentially Private Synthetic Text Improve Capabilities?

Peihan Liu*, Lucas Rosenblatt, Weiwei Kong, Natalia Ponomareva, Gautam Kamath, Rachel Cummings, Roxana Geambasu, Yu Gan, Lillian Tsai, Alex Bie* (Contribution order)

Manuscript

Robust Statistical Estimators with Bounded Empirical Sensitivity

Valentio Iverson, Gautam Kamath, Argyris Mouzakis, Adam Smith

Manuscript

DriftXpress: Faster Drifting Models via Projected RKHS Fields

Ali Falahati, Elliot Creager*, Gautam Kamath*, Shubhankar Mohapatra* (Contribution order)

Manuscript

Are Targeted Data Poisoning Attacks as Effective as We Think?

William Xu*, Chenyu Zhang*, Yihan Wang, Matthew Y. R. Yang, Zuoqiu Liu, Yaoliang Yu, Gautam Kamath, Yiwei Lu (Contribution order)

Manuscript

Report of the 1st Workshop on Generative AI and Law

A. Feder Cooper, Katherine Lee, James Grimmelman, Daphne Ippolito, Christopher Callison-Burch, Christopher A. Choquette-Choo, Niloofar Mireshghallah, Miles Brundage, David Mimno, Madiha Zahrah Choksi, Jack M. Balkin, Nicholas Carlini, Christopher De Sa, Jonathan Frankle, Deep Ganguli, Bryant Gipson, Andres Guadamuz, Swee Leng Harris, Abigail Z. Jacobs, Elizabeth Joh, Gautam Kamath, Mark Lemley, Cass Matthews, Christine McLeavey, Corynne McSherry, Milad Nasr, Paul Ohm, Adam Roberts, Tom Rubin, Pamela Samuelson, Ludwig Schubert, Kristen Vaccaro, Luis Villa, Felix Wu, Elana Zeide

Manuscript

A Primer on Private Statistics

Gautam Kamath, Jonathan Ullman
Manuscript

Bounds on the Expectation of the Maximum of Samples from a Gaussian
Gautam Kamath
Manuscript

SELECTED TALKS

Kempner Institute Seminar Series TBD Cambridge, Massachusetts, USA	April 2027
46th IARCS Annual Conference on Foundations of Software Technology and Theoretical Computer Science (FSTTCS 2026) TBD (Plenary talk) New Delhi, India	December 2026
Rutgers/DIMACS Theory of Computing Seminar TBD New Brunswick, New Jersey, USA	November 2026
Shonan Meeting on Exploring Responsible AI through Security and Privacy TBD Tokyo, Japan	October 2026
Princeton University Theory Lunch TBD Princeton, New Jersey, USA	September 2026
Workshop on the Intersections of Differential Privacy and Sublinear Algorithms Optimal Differentially Private Sampling of Unbounded Gaussians Chicago, Illinois, USA	July 2026
53rd EATCS International Colloquium on Automata, Languages, and Programming (ICALP 2026) Private Estimation via Robustness (Presburger Lecture) Egham, UK	July 2026
KAIST @ ICML 2026 Career Advice for Junior Researchers & Students (Panel) Seoul, South Korea	July 2026
Forty-Third International Conference on Machine Learning (ICML 2026) Unlearning Data at Scale (Tutorial) Seoul, South Korea	July 2026
AICan 2026 Peer Review at the Breaking Point; Do We Have Solutions? (Panel) Victoria, British Columbia, Canada	June 2026
Columbia University Workshop on Robust Statistics Statistical Estimators with Bounded Empirical Sensitivity New York, New York, USA	April 2026
Apple Privacy-Preserving Machine Learning & AI Workshop Demystifying Foreground-Background Memorization in Diffusion Models Virtual	March 2026
University of Waterloo Statistics and Biostatistics Seminar The Broader Landscape of Robustness in Algorithmic Statistics Waterloo, Ontario, Canada	March 2026
Remarkable 2026 Privacy and Security Workshop Membership Inference Attacks Cannot Prove that a Model Was Trained On Your Data Toronto, Ontario, Canada	February 2026
Imperial College London Data Science Institute Seminar When Can we Remove Data from Trained Models? London, UK	January 2026
Laboratory for AI Security Research Workshop When Can we Remove Data from Trained Models?	January 2026

London, UK	
2025 Daegu Global Robot Business Forum	October 2025
Protecting Personal Data in AI Model Training	
Daegu, South Korea	
ETH AI Center Academic Talk Series	October 2025
Demystifying Foreground-Background Memorization in Diffusion Models	
Zurich, Switzerland	
Symposium on Mathematical Foundations of Trustworthy Machine Learning	October 2025
The Broader Landscape of Robustness in Algorithmic Statistics	
Ascona, Switzerland	
The 7th Privacy-Preserving Machine Learning Workshop 2025	August 2025
The Broader Landscape of Robustness in Algorithmic Statistics	
Santa Barbara, California, USA	
3rd Vector Machine Learning Security and Privacy Workshop	July 2025
Machine Unlearning Fails to Remove Data Poisoning Attacks	
Toronto, Ontario, Canada	
ELSA Workshop on Privacy-Preserving Machine Learning	March 2025
The Promise and Pitfalls of Public Data in Private ML	
Panel: Practice of Differential Privacy	
Bertinoro, Italy	
ACL Year-Round Mentorship	January 2025
Panel: How to broadcast your research to a wider audience?	
Virtual	
2nd Southern Ontario Learning Theory Workshop	November 2024
The Broader Landscape of Robustness in Mean Estimation	
Waterloo, Ontario, Canada	
Vector Institute ICML and ICLR 2024 Conference Highlights	November 2024
Considerations for Differentially Private Learning with Large-Scale Public Pretraining	
Virtual	
Charles River Privacy Day 2024	November 2024
The Broader Landscape of Robustness in Mean Estimation	
Boston, Massachusetts, USA	
National University of Singapore AI Lab Horizons Series	October 2024
The Promise and Pitfalls of Public Pre-training in Differentially Private Machine Learning	
Virtual	
IEEE Computer Society Santa Clara Valley Chapter IEEE Day Event	October 2024
AMA on Data Privacy, Machine Unlearning, and More	
Virtual	
Google Core Labs Seminars	September 2024
The Promise and Pitfalls of Public Pre-training in Differentially Private Machine Learning	
Virtual	
8th Annual Toronto Machine Learning Summit 2024	July 2024
Privacy Risks and Protections in Machine Learning Systems (Opening Keynote)	
Toronto, Ontario, Canada	
2024 Canadian Computing Olympiad	June 2024
Selection, with Rounds or Adversarial Comparisons	
Waterloo, Ontario, Canada	
Yale CSPC 471/571: Trustworthy Deep Learning	April 2024
Differentially Private Machine Learning (Guest Lecture)	
Virtual	
New York University Courant CS Colloquium	March 2024
Principled Approaches for Trustworthy Algorithms, Statistics, and Machine Learning	
New York, New York, USA	
Cornell University ORIE Colloquium	March 2024
Principled Approaches for Trustworthy Algorithms, Statistics, and Machine Learning	
Ithaca, New York, USA	
Yale University CS Talk	February 2024

Principled Approaches for Trustworthy Algorithms, Statistics, and Machine Learning New Haven, Connecticut, USA	
UC San Diego HDSI Seminar Principled Approaches for Trustworthy Algorithms, Statistics, and Machine Learning La Jolla, California, USA	February 2024
Vector Institute Distinguished Talk Series The Promise and Pitfalls of Public Data in Private ML Virtual	November 2023
Vector Institute Machine Learning Theory Workshop Statistical Estimation with Privacy Constraints Waterloo, Ontario, Canada	November 2023
University of Guelph CARE-AI Seminar Protecting Individual Privacy in Machine Learning Guelph, Ontario, Canada	November 2023
Columbia University Robust Statistics and Privacy Workshop Robust Estimators for Private Estimation New York, New York, USA	October 2023
6th Eastern Great Lakes (EaGL) Theory of Computation Workshop Differentially Private Mean Estimation Rochester, New York, USA	October 2023
JSM 2023 Invited Session on Robust Statistics and Differential Privacy Private Estimators from Robust Statistics Toronto, Ontario, Canada	August 2023
BIRS Workshop on Contextual Integrity for Differential Privacy Considerations for Differentially Private Learning with Large-Scale Public Pretraining Okanagan, British Columbia, Canada	July 2023
ICML 2023 Workshop on Generative AI + Law What does Differential Privacy have to do with Copyright? Honolulu, Hawaii, USA	July 2023
ICML 2023 Black in AI Social AI and Society (Panel member) Honolulu, Hawaii, USA	July 2023
Vector Machine Learning Security and Privacy Workshop The Promise and Pitfalls of Public Data in Private ML Toronto, Ontario, Canada	July 2023
IISC Bangalore Joint Telematics Group Summer School on Information Theory Introduction to Differential Privacy (Invited Tutorial, 8 hours of lectures) Bangalore, India	June 2023
Upper Bound Workshop Bridge the Gap: Differential Privacy and Statistical Analysis Recent Connections between Differential Privacy and Robustness Edmonton, Alberta, Canada	May 2023
Cybersecurity, Privacy, and AI in Health Data: Advancements and Challenges An Introduction to Differential Privacy for Analysis of Sensitive Data Ottawa, Ontario, Canada	May 2023
Vector Faculty Research Meeting Public Data for Private Machine Learning Virtual	April 2023
Waterloo CPI Conference The Weaponization of Disinformation in Canada Problematic Disinformation (Panel member) Waterloo, Ontario, Canada	April 2023
Waterloo Computer Science Club Prof Talk Differential Privacy in Machine Learning Waterloo, Ontario, Canada	March 2023
Apple Workshop on Privacy Preserving ML Rethinking Benchmarks for Private Image Classification Virtual	February 2023
KAUST Rising Stars in AI Symposium 2023	February 2023

Differentially Private Fine-tuning of Language Models Thuwal, Saudi Arabia	
IEEE Conference on Secure and Trustworthy Machine Learning An Introduction to Differential Privacy (Invited Tutorial) Raleigh, North Carolina, USA	February 2023
Georgia Tech AI4OPT Seminar Efficient Private Mean Estimation Atlanta, Georgia, USA	February 2023
IMS International Conference on Statistics and Data Science The Role of (Statistical) Bias in Private Estimation Florence, Italy	December 2022
Cornell Computer Science Theory Seminar Efficient Private Mean Estimation Ithaca, New York, USA	December 2022
UC Berkeley BLISS Seminar Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism Virtual	November 2022
US Census ML/AI Discussion Group CoinPress: Practical Private Point Estimation and Confidence Intervals Virtual	November 2022
LinkedIn Data Tech Talk Series Differentially Private Fine-tuning of Language Models Virtual	October 2022
Canadian AI Federated Learning Workshop Differentially Private Fine-tuning of Language Models Toronto, Ontario, Canada	October 2022
Rutgers Business School MSIS Department Seminar Series Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism Virtual	October 2022
Columbia Statistics Seminar Series CoinPress: Practical Private Point Estimation and Confidence Intervals New York, New York, USA	October 2022
University of British Columbia CAIDA Seminar Series Differentially Private Fine-tuning of Language Models Vancouver, British Columbia, Canada	August 2022
PIMS Mathematics of Ethical Decision-making Systems Seminar Statistical Estimation with Differential Privacy Victoria, British Columbia, Canada	August 2022
Meta Lunch and Learn Statistical Estimation with Differential Privacy Virtual	August 2022
Fields Workshop on Differential Privacy and Statistical Data Analysis Premonitions of Public Data for Private ML Toronto, Ontario, Canada	July 2022
International Conference on Robust Statistics Robust Estimation for Random Graphs Waterloo, Ontario, Canada	July 2022
Google Privacy Seminar Differentially Private Fine-tuning of Language Models Virtual	June 2022
University of Washington Theory Seminar Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism Virtual	April 2022
University of Waterloo Probability Seminar Series Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism Waterloo, Ontario, Canada	April 2022
Apple Workshop on Privacy Preserving ML	April 2022

Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism	
Virtual	
Apple Workshop on Privacy Preserving ML	April 2022
Differentially Private Fine-tuning of Language Models	
Virtual	
Simons Institute Data Privacy: Foundations and Applications Reunion	March 2022
Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism	
Virtual	
UMass Amherst Machine Learning & Friends Lunch	February 2022
Differentially Private Fine-tuning of Language Models	
Virtual	
BIRS Workshop on Mathematical Statistics and Learning	November 2021
Efficient Mean Estimation with Pure Differential Privacy via a Sum-of-Squares Exponential Mechanism	
Banff, Alberta, Canada	
Google Workshop on Federated Learning and Analytics	November 2021
Differentially Private Fine-tuning of Language Models	
Virtual	
ML Collective Deep Learning: Classics and Trends	November 2021
Differentially Private Fine-tuning of Language Models	
Virtual	
IDEAL Mini-workshop on New Direction on Robustness in ML	November 2021
Statistical Estimation with Differential Privacy	
Virtual	
London Symposium on Information Theory	May 2021
Private Hypothesis Selection	
Virtual	
TrustML Seminar	April 2021
CoinPress: Practical Private Estimation	
Virtual	
Boston-Area DP Seminar	April 2021
Hypothesis Selection with Privacy	
Virtual	
Virtual Conference on Robustness and Privacy	March 2021
Differentially Private Mean and Covariance Estimation	
Virtual	
Google Privacy and Machine Learning Seminar	March 2021
CoinPress: Practical Private Mean and Covariance Estimation	
Virtual	
McGill Statistics Seminar	February 2021
CoinPress: Practical Private Point Estimation and Confidence Intervals	
Virtual	
University of Waterloo ML + Logic Seminar	January 2021
Robustness in Unsupervised and Supervised Machine Learning	
Virtual	
Simons Institute Reading Group	November 2020
Differentially Private Statistical Estimation	
Virtual	
University of Toronto Theory Seminar	October 2020
Hypothesis Selection with Privacy Constraints	
Virtual	
University of Pennsylvania Wharton Statistics Seminar	September 2020
CoinPress: Practical Private Point Estimation and Confidence Intervals	
Virtual	
Northwestern University IDEAL Seminar	August 2020
Theory and Practice for Private Statistical Estimation - Gaussians and Beyond	
Virtual	
Harvard Privacy Tools Group Meeting	August 2020

CoinPress: Practical Private Mean and Covariance Estimation Virtual	
Joint Statistical Meetings Differentially Private Mean and Covariance Estimation Virtual	August 2020
Carnegie Mellon University Theory Lunch Hypothesis Selection with Privacy Constraints Virtual	August 2020
University of Waterloo Algorithms and Complexity Seminar Robustness in Unsupervised and Supervised Machine Learning Virtual	May 2020
Google Mountain View Algorithms Group Meeting Privately Learning High-Dimensional Distributions Mountain View, California, USA	February 2020
Information Theory and Applications Workshop Private Hypothesis Selection San Diego, California, USA	February 2020
Stanford University Management Science & Engineering Seminar Principled Tools for Modern Statistical Challenges Stanford, California, USA	February 2020
National Technical University of Athens Corelab Seminar Privately Learning High-Dimensional Distributions Athens, Greece	July 2019
Workshop on Algorithms for Learning and Economics Efficient Multivariate Robust Statistics Rhodes, Greece	July 2019
Google Seattle Cerebra Journal Club Estimating a Gaussian: Robustly or Privately Seattle, Washington, USA	April 2019
Simons Institute Workshop on Data Privacy: From Foundations to Applications Privately Learning High-Dimensional Distributions Berkeley, California, USA	March 2019
Berkeley Theory Lunch Privately Learning High-Dimensional Distributions Berkeley, California, USA	March 2019
MIT Algorithms and Complexity Seminar Privately Learning High-Dimensional Distributions Cambridge, Massachusetts, USA	February 2019
Berkeley BLISS Seminar Privately Learning High-Dimensional Distributions Berkeley, California, USA	February 2019
Information Theory and Applications Workshop Privately Learning High-Dimensional Distributions San Diego, California, USA	February 2019
Caltech Mathematics of Information Seminar Privately Learning High-Dimensional Distributions Pasadena, California, USA	January 2019
Simons Institute Data Privacy: Foundations and Applications Boot Camp Statistical Inference and Privacy Berkeley, California, USA	January 2019
Symposium on Discrete Algorithms Anaconda: A Non-Adaptive Conditional Sampling Algorithm for Distribution Testing San Diego, California, USA	January 2019
Microsoft Research Machine Learning and Optimization Lunch Privately Learning High-Dimensional Distributions Redmond, Washington, USA	November 2018
Simons Institute Workshop on Robust and High-Dimensional Statistics	November 2018

Realizing Robustness Berkeley, California, USA	
TTIC Workshop on Computational Efficiency and High-Dimensional Robust Statistics	August 2018
Beyond Theory: Realizing Robustness Chicago, Illinois, USA	
BIRS Mathematical Foundations of Data Privacy Workshop	May 2018
Differentially Private Hypothesis Testing and Property Estimation Banff, Alberta, Canada	
CRM Modern Challenges of Learning Theory Workshop	April 2018
Robustness in Unsupervised and Supervised Machine Learning Montreal, Quebec, Canada	
MIT LIDS and Stats Tea	April 2018
INSPECTRE: Privately Estimating the Unseen Cambridge, Massachusetts, USA	
Conference on Information Sciences and Systems	March 2018
Hypothesis Testing with Alternative Distances Princeton, New Jersey, USA	
Boston University Computer Science Seminar	February 2018
Principled Tools for Modern Statistical Data Science Boston, Massachusetts, USA	
McGill University Computer Science Seminar	February 2018
Principled Tools for Modern Statistical Data Science Montreal, Quebec, Canada	
University of Waterloo Computer Science Seminar	February 2018
Principled Tools for Modern Statistical Data Science Waterloo, Ontario, Canada	
Symposium on Discrete Algorithms	January 2018
Which Distribution Distances are Sublinearly Testable? New Orleans, Louisiana, USA	
University of Pennsylvania Theory Seminar	December 2017
Statistical Hypothesis Testing in the Modern Age Philadelphia, Pennsylvania, USA	
Boston University Algorithms and Theory Seminar	November 2017
Statistical Hypothesis Testing in the Modern Age Boston, Massachusetts, USA	
University of Massachusetts Amherst Theory Seminar	October 2017
Statistical Hypothesis Testing in the Modern Age Amherst, Massachusetts, USA	
McMaster Seminar in Computing and Software	October 2017
Statistical Hypothesis Testing in the Modern Age Hamilton, Ontario, Canada	
FOCS Workshop on Frontiers in Distribution Testing	October 2017
Testing with Alternative Distances Berkeley, California, USA	
Cornell Theory Lunch	September 2017
Robust Estimators in High Dimensions without the Computational Intractability Ithaca, New York, USA	
Cornell Theory Seminar	September 2017
Statistical Hypothesis Testing in the Modern Age Ithaca, New York, USA	
International Conference on Machine Learning	August 2017
Priv'IT: Private and Sample Efficient Identity Testing Sydney, Australia	
ICML Workshop on Private and Secure Machine Learning	August 2017
Priv'IT: Private and Sample Efficient Identity Testing Sydney, Australia	
Northeastern Theory Seminar	March 2017

Some Frontiers in Distribution Testing Boston, Massachusetts, USA	
University of Pennsylvania Theory Lunch Optimal Testing for Properties of Distributions Philadelphia, Pennsylvania, USA	September 2016
Symposium on Theory of Computing A Size-Free CLT for Poisson Multinomials and its Applications Cambridge, Massachusetts, USA	June 2016
MIT Signals, Information, and Algorithms Laboratory Group Meeting Optimal Testing for Properties of Distributions Cambridge, Massachusetts, USA	March 2016
University of Massachusetts Boston Computer Science Seminar Optimal Testing for Properties of Distributions Boston, Massachusetts, USA	February 2016
Berkeley Theory Lunch Optimal Testing for Properties of Distributions Berkeley, California, USA	September 2015
Conference on Learning Theory Faster and Sample Near-Optimal Algorithms for Proper Learning Mixtures of Gaussians Barcelona, Spain	June 2014
Interdisciplinary Workshop on Information and Decision in Social Networks An Analysis of One-Dimensional Schelling Segregation Cambridge, Massachusetts, USA	November 2012
Symposium on Theory of Computing An Analysis of One-Dimensional Schelling Segregation New York, New York, USA	May 2012
Winner of Best Student Presentation Award	

GRADUATE STUDENTS

Matthew Regehr (Fall 2023 - Present, co-advised with Shai Ben-David)
 PhD, Computer Science, University of Waterloo
 Awarded **Waterloo CPI Cybersecurity and Privacy Excellence Graduate Scholarship**
 Awarded **Ontario Graduate Scholarship**
 Awarded **NSERC Canada Graduate Scholarship - Doctorate**

Argyris Mouzakis (Fall 2020 - Summer 2026)
 PhD, Computer Science, University of Waterloo
 Thesis: The Sample Complexity of Differentially Private Statistical Estimation
 Next Position: Postdoc at Institute for Foundations of Machine Learning (IFML), UT Austin
 Awarded **Onassis Foundation Scholarship**
 Awarded **Cheriton Scholarship (Type I)**

Jimmy Z. Di (Fall 2023 - Summer 2025)
 MMath, Computer Science, University of Waterloo
 Thesis: Demystifying Foreground-Background Memorization in Diffusion Models
 Next Position: PhD in Computer Science at University of Wisconsin, Madison
 Awarded **Vector Scholarship in Artificial Intelligence**
 Awarded **Ontario Graduate Scholarship**

Sabrina Mokhtari (Fall 2022 - Summer 2024)
 MMath, Computer Science, University of Waterloo
 Thesis: Revisiting Benchmarks for Privacy-Preserving Image Classification
 Next Position: Machine Learning Associate at Vector Institute
 Current Position: Member of Technical Staff at Oracle
 Awarded **Vector Scholarship in Artificial Intelligence**
 Awarded **NSERC Canada Graduate Scholarship - Master's**

Sara Kodeiri (Fall 2022 - Summer 2024)

MMath, Computer Science, University of Waterloo

Thesis: Towards Standardized Evaluation in Differentially Private Image Classification: A Critical Approach

Next Position: Machine Learning Associate at Vector Institute

Current Position: Associate Applied ML Specialist at Vector Institute

Matthew Regehr (Fall 2021 - Summer 2023, co-advised with Shai Ben-David)

MMath, Computer Science, University of Waterloo

Thesis: A Bias-Variance-Privacy Trilemma for Statistical Estimation

Next Position: PhD in Computer Science at University of Waterloo

Awarded **Vector Scholarship in Artificial Intelligence**

Awarded **NSERC Canada Graduate Scholarship - Master's**

Yaxian Alex Bie (Fall 2021 - Summer 2023, co-advised with Shai Ben-David)

MMath, Computer Science, University of Waterloo

Thesis: Private Distribution Learning with Public Data

Current Position: Research Scientist at Google Research

Awarded **Vector Scholarship in Artificial Intelligence**

Awarded **Ontario Graduate Scholarship**

Mahbod Majid (Fall 2020 - Fall 2022)

MMath, Computer Science, University of Waterloo

Thesis: Efficient and Differentially Private Statistical Estimation via a Sum-of-Squares Exponential Mechanism

Next Position: PhD in Machine Learning at Carnegie Mellon University

Current Position: PhD in Applied Mathematics at MIT

Awarded **Waterloo CPI Cybersecurity and Privacy Excellence Graduate Scholarship**

Awarded **Faculty of Mathematics Graduate Research Excellence Award**

Awarded **University Finalist for the Governor General's Gold Medal**

Awarded **University Finalist for the Alumni Gold Medal**

Christian Covington (Fall 2020 - Summer 2022, co-advised with Xi He)

MMath, Computer Science, University of Waterloo

Thesis: Unbiased Statistical Estimation and Valid Confidence Intervals Under Differential Privacy

Next Position: PhD in Biostatistics at Harvard University

POSTDOCS

Jacob Imola (Spring 2026 - Present)

Vikrant Singhal (Fall 2021 - Fall 2023)

Next Position: Research Associate at OpenDP

Current Position: (fixed-term) Assistant Professor at University of Copenhagen

UNDERGRADUATE RESEARCH ADVISING

Jack Sanderson (Fall 2025 - Present)

Valentio Iverson (Fall 2023 - Summer 2025)

Next position: PhD in EECS at MIT

Published "Optimal Differentially Private Sampling of Unbounded Gaussians" in COLT 2025

Awarded **CRA Outstanding Undergraduate Researcher, Honorable Mention**

Awarded **Germain – Erdős Undergraduate Award in Mathematics Research**

Matthew Y.R. Yang (Fall 2022 - Summer 2024)

Next position: MS in Machine Learning at Carnegie Mellon University

Published "Indiscriminate Data Poisoning Attacks on Pre-trained Feature Extractors" in SaTML 2024

Published "Disguised Copyright Infringement of Latent Diffusion Models" in ICML 2024

Awarded **CRA Outstanding Undergraduate Researcher, Finalist**

Chris Trevisan (Spring 2022 - Spring 2024)

Published “Sorting and Selection in Rounds with Adversarial Comparisons” in SODA 2024

Awarded **CRA Outstanding Undergraduate Researcher, Awardee**

Jimmy Z. Di (Fall 2021 - Summer 2023)

Next position: MMath in Computer Science at University of Waterloo

Current position: PhD in Computer Science at University of Wisconsin, Madison

Published “Hidden Poison: Machine Unlearning Enables Camouflaged Poisoning Attacks” in NeurIPS 2023

Ruiyun Chao (Fall 2022)

Olivia Ma (Fall 2022)

Next position: Master of Science in Computing (AI & ML) at Imperial College London

Valerie Liu (Fall 2022)

Next position: Master’s student in Computing Science at University of Alberta

Jack Douglas (Summer 2022)

Next position: Software Engineer at Databricks

Published “Hidden Poison: Machine Unlearning Enables Camouflaged Poisoning Attacks” in NeurIPS 2023

Andrew Guo (Summer 2022)

Chirag Jindal (Summer 2022)

Landy Xu (Spring 2021 - Fall 2021)

Next position: Master of Science in Applied Computing at University of Toronto

Xingtu Liu (Fall 2020 - Summer 2021)

Next position: Master of Science in Computer Science at Simon Fraser University

Current position: PhD in Statistics and Data Science at University of California, Los Angeles

Published “Improved Rates for Differentially Private Stochastic Convex Optimization with Heavy-Tailed Data” in ICML 2022

Nicholas Vadivelu (Fall 2020 - Summer 2021)

Next position: Quantitative Research and Data Scientist at Citadel

Awarded **CRA Outstanding Undergraduate Researcher, Runner-Up**

Awarded **Jessie W.H. Zou Memorial Award**

Published “Enabling Fast Differentially Private SGD via Just-in-Time Compilation and Vectorization” in NeurIPS 2021

Pranav Subramani (Fall 2019 - Summer 2021)

Next position: Quantitative Researcher at Cubist Systematic Strategies

Published “Enabling Fast Differentially Private SGD via Just-in-Time Compilation and Vectorization” in NeurIPS 2021

Sourav Biswas (Fall 2019 - Summer 2021)

Next position: Researcher at Waabi and PhD in Computer Science at University of Toronto

Awarded **CRA Outstanding Undergraduate Researcher, Honorable Mention**

Published “CoinPress: Practical Private Mean and Covariance Estimation” in NeurIPS 2020

OTHER MENTORSHIP

Shubhankar Mohapatra (Fall 2019 - Spring 2026)

Next position: Senior Scientist at A*STAR

Published “The Role of Adaptive Optimizers for Honest Private Hyperparameter Selection” in AAAI 2022

Published “Rethinking Benchmarks for Differentially Private Image Classification” in IEEE Data Engineering Bulletin

Yiwei Lu (Summer 2020 - Summer 2025)

Next position: Assistant Professor in Engineering Design and Teaching Innovation at University of Ottawa

Published “Indiscriminate Data Poisoning Attacks on Neural Networks” in TMLR, 2022

Published “Exploring the Limits of Model-Targeted Indiscriminate Data Poisoning Attacks” in ICML 2023

Published “Indiscriminate Data Poisoning Attacks on Pre-trained Feature Extractors” in SaTML 2024

Published “Disguised Copyright Infringement of Latent Diffusion Models” in ICML 2024

Published “Machine Unlearning Fails to Remove Data Poisoning Attacks” in ICLR 2025

Ishaq Aden-Ali (Summer 2020 - Summer 2021)

Next position: PhD in Computer Science at UC Berkeley

Published “On the Sample Complexity of Privately Learning Unbounded High-Dimensional Gaussians” in ALT 2021

Sushant Agarwal (Spring 2022 - Summer 2022)

Next position: PhD in Computer Science at Northeastern University

SELECTED PRESS COVERAGE AND QUOTES

Is it Possible to Erase Digital Memory?

Communications of the ACM, April 2026

Should you trust ChatGPT and DeepSeek with your data? Here’s what the experts say

Toronto Star, April 2025

Teens Are Doing AI Research Now. Is That a Good Thing?

The Chronicle of Higher Education, January 2025

“Copyright traps” could tell writers if an AI has scraped their work

MIT Technology Review, July 2024

Artists Are Slipping Anti-AI ‘Poison’ into Their Art. Here’s How It Works

Scientific American, March 2024

The modern Luddites’ fight for the human premium in AI

Canvas8 Report, February 2024

1 big thing: It can be hard for AI to forget

Axios Science, January 2024

New tools help artists fight AI by directly disrupting the systems

NPR All Things Considered, November 2023

This new data poisoning tool lets artists fight back against generative AI

MIT Technology Review, October 2023

Police in Essex County have started using licence plate scanners. Here’s how they work

CBC, April 2023

Who Is Working to End the Threat of AI-Generated Deepfakes, and Why Is It So Difficult?

Gizmodo, November 2022

Can AI Learn to Forget?

Communications of the ACM, April 2022

Now That Machines Can Learn, Can They Unlearn?

Wired Magazine, August 2021

GRANTS

Ontario Early Researcher Award

Sole PI

04/2024 - 04/2029

Canada CIFAR AI Chair

Sole PI

03/2023 - 03/2028

Ontario Research Fund: Research Infrastructure

Co-PI, with Xi He

11/2022 - 11/2024

Apple Unrestricted Gift

Sole PI

8/2022

Compute Canada Resources for Research Groups

Co-PI, with Xi He

4/2022 - 3/2023

Google Unrestricted Gift

Sole PI

2/2022

Canada Foundation for Innovation John R. Evans Leaders Fund

Co-PI, with Xi He

8/2021 - 1/2024

NSERC Discovery Grant

Sole PI

4/2020 - 3/2027

NSERC Discovery Grant - Accelerator Supplement

Sole PI

4/2020 - 3/2023

NSERC Discovery Grant - Launch Supplement

Sole PI

4/2020 - 3/2021

Compute Canada Resources for Research Groups

Co-PI, with Xi He

4/2021 - 3/2022

Compute Canada Resources for Research Groups

Co-PI, with Xi He

4/2020 - 3/2021

University of Waterloo Startup Grant

Sole PI

7/2019 - 6/2024

ADDITIONAL HONOURS AND AWARDS

Senior Member, IEEE	December 2023
Notable Reviewer, SaTML 2023	February 2023
Best Reviewer Award, CCS 2021	November 2021
Top Graduate Instructor for CS 761 in Fall 2019	March 2020
Top 5% Highest-Scoring Reviewer for ICML 2019	June 2019
Top 30% Highest-Scoring Reviewer for NeurIPS 2018	December 2018
MIT Akamai Presidential Graduate Fellowship	September 2012 - May 2013
Best Student Presentation Award, STOC 2012	May 2012
Cornell Computer Science Prize for Academic Excellence	May 2012
Eight time Dean's list at Cornell University	Fall 2008 - Spring 2012
Recognized by Cornell CS for outstanding work as TA for CS 3110 and CS 4820	Spring 2012
John G. Pertsch Jr. Prize for second highest GPA in Cornell ECE	Spring 2011
Recognized by Cornell CS for outstanding work as TA for CS 1114	Spring 2010
Canadian Open Mathematics Challenge Gold Medalist in Central Ontario Region	Spring 2007

TEACHING

Instructor	New York University	2026 -
CSCI-GA 3033: Honors Machine Learning		
Fall 2026		
Instructor	University of Waterloo	2019 - 2026
CS 240: Data Structures and Data Management		
Spring 2020		
CS 480: Introduction to Machine Learning		
Winter 2021 (2 sections), Spring 2021 (2 sections), Spring 2022 (2 sections),		
Fall 2023 (2 sections), Fall 2025 (2 sections)		
CS 761: Randomized Algorithms		
Fall 2019		
CS 860: Algorithms for Private Data Analysis		
Fall 2020, Fall 2022, Winter 2026		
Teaching Assistant	Massachusetts Institute of Technology	2015-2017
6.853: Algorithmic Game Theory and Data Science		
Spring 2017		
6.856: Randomized Algorithms		
Spring 2015		
Teaching Assistant	Cornell University	2010 - 2012
CS 1114: Intro to Computing with Matlab and Robotics		
Spring 2010, Spring 2012		
CS 2850: Networks		
Fall 2011		
CS 3110: Data Structures and Functional Programming		
Summer 2010, Fall 2010, Spring 2011, Fall 2011, Spring 2012		
CS 4820: Introduction to Algorithms		
Summer 2010, Spring 2011, Spring 2012		

PROFESSIONAL ACTIVITIES

Journal Editor-in-Chief: TMLR (July 2023 - December 2026)
Conference Program Committee Chair: ALT 2025
Conference General Chair: STOC 2020
Workshop Program Chair: TPDP 2021, TPDP 2022, UpML 2022, Re-Data 2026
Senior Area Chair: NeurIPS 2026

Conference Core Program Committee or Area Chair: SODA 2020, ICML 2020, ICALP 2020, RANDOM 2020, ALT 2021, ICLR 2021, FORC 2021, COLT 2021, CCS 2021, NeurIPS 2021, ESA 2021, ALT 2022, SODA 2022, ICLR 2022, AAAI 2022, COLT 2022, NeurIPS 2022, SaTML 2023, FOCS 2023, ICLR 2023, ALT 2023, USENIX Security 2023, COLT 2023, FAccT 2023, ICML 2023, NeurIPS 2023, COLT 2024, NeurIPS 2024, STOC 2025, COLT 2025, ICML 2025 Position Paper Track, SODA 2026, SaTML 2026, ALT 2026, COLT 2026, STOC 2027, ALT 2027

Machine Learning Conference Program Committee Member (i.e., reviewer): NIPS 2016, ICML 2018, NeurIPS 2018, AISTATS 2019, ICML 2019, NeurIPS 2019, AAAI 2020, AISTATS 2020, FAccT 2021, UAI 2022, CANAI 2024

Workshop Program Committee Member: TPDP 2019, PriML 2019, TPDP 2020, PPML 2020, PriML 2021, ICBINB 2021, ICBINB 2022, ICBINB 2023, Regulatable ML 2023, PPAI 2024, TPDP 2024, TPDP 2025

Journal Guest Editor: TALG Special Issue for SODA 2020

Journal Action Editor: Transactions on Machine Learning Research (March 2022 - July 2023), Journal of Privacy and Confidentiality (April 2025 -)

Other: ICML 2021 Workshop Reviewer, NeurIPS 2021 Social Chair, NeurIPS 2022 Ethics Reviewer, ICML 2023 Workshop Reviewer, SaTML 2024 Social Media Chair, NeurIPS 2023 Ethics Reviewer, TMLR 2022 Outstanding Paper Committee, ICML 2024 Workshop Reviewer, 2024 Caspar Bowden Award for Outstanding Research in Privacy Enhancing Technologies Committee, SaTML 2025 Social Media Chair, SaTML 2026 Social Media Chair, AAAI 2026 Social Media and Publicity Chair, ICML 2026 Communications Co-Chair, ICLR 2026 Best Paper Award Committee (Chair), SaTML 2027 Social Media Chair

Conference external reviewer: AAAI, AISTATS, ALT, COLT, FAccT, FOCS, ICALP, ICML, ICML Workshops, ISAAC, ISIT, ITCS, NeurIPS, RANDOM, SODA, STACS, STOC

Journal reviewer: Algorithmica, Annals of Statistics, Behavior Research Methods, Foundations and Trends in Theoretical Computer Science, Foundations of Data Science, Journal of Machine Learning Research, Journal of Privacy and Confidentiality, Statistica Sinica, Theory of Computing Systems

Grant reviewer: Natural Sciences and Engineering Research Council, Blavatnik Interdisciplinary Cyber Research Centre, National Science Foundation, Singapore Ministry of Education, Swiss National Science Foundation, National Science Center Poland

Member of external review committee for Reed College Computer Science Department (2025)

Co-Organizer of COLM 2026 Workshop “Workshop on Responsibly Enabling Data for Foundation Models” (October 2026)

Organizer of Vector Institute Foundations of AI Security Symposium 2026 (March 2026)

Co-organizer of NeurIPS 2025 Social “Learning Theory Alliance Social” (December 2025)

Organizer of Vector Institute Workshop “Vector Machine Learning Security and Privacy Workshop” (July 2025)

Organizer of Vector Institute Workshop “Vector Machine Learning Security and Privacy Workshop” (July 2024)

Co-organizer of TTIC Workshop “New Frontiers in Robust Statistics” (June 2024)

Organizer of Vector Institute Workshop “Vector Machine Learning Security and Privacy Workshop” (July 2023)

Lead organizer of Fields Institute Workshop “Workshop on Differential Privacy and Statistical Data Analysis” (June 2022)

Co-organizer of ICML 2022 Workshop “Updatable Machine Learning” (July 2022)

Co-organizer of ICML 2022 Workshop “Theory and Practice of Differential Privacy” (July 2022)

Co-organizer of ICML 2021 Workshop “Theory and Practice of Differential Privacy” (July 2021)

Co-organizer of ICLR 2021 Workshop “Distributed and Private Machine Learning” (May 2021)

Co-organizer of NeurIPS 2020 Social “Data Privacy: Academia, Industry, Policy, and Society” (December 2020)

Co-organizer of FOCS 2019 Workshop “A TCS Quiver” (November 2019)

Co-organizer of FOCS 2017 Workshop “Frontiers in Distribution Testing” (October 2017)

Co-organizer of FOCS 2016 Workshop “Orthogonal Polynomials and Applications” (October 2016)

Organizer of the Second Annual Sublinear Algorithms and Big Data Day (April 2015)

Cofounder and organizer of MIT Theory Lunch (Fall 2012 - Summer 2013)

Advisor for Danny Lewin MIT Theory Student Retreat (Fall 2014, 2016, 2017)

Organizer of Second Annual Danny Lewin MIT Theory Student Retreat (October 2013)

IEEE Conference on Secure and Trustworthy Machine Learning, Steering Committee (July 2026 - Present)

Symposium on Foundations of Responsible Computing, Executive Committee (November 2025 - Present)

International Conference on Machine Learning, Board Member (July 2025 - Present)

Association for Computational Learning, Director (July 2024 - Present)

Association for Algorithmic Learning Theory, Steering Committee (June 2024 - February 2029)

Executive Committee of Learning Theory Alliance (November 2023 - Present)

Steering Committee of Theory and Practice of Differential Privacy (October 2023 - Present)

Founder and co-organizer of DifferentialPrivacy.org (July 2020 - Present)

Editor of ALT Highlights (April 2021 - July 2021)

Maintainer of CS Theory Blog Aggregator (January 2019 - Present)

Co-organizer for the TCS+ online seminar series in Theoretical Computer Science (August 2014 - Present)

Editor of Property Testing Review (March 2016 - June 2020)

Editor of MIT Theory of Computation Student Blog (November 2013 - October 2016)

Co-organizer of New York University CS Theory Seminar (August 2026 - Present)

Member of New York University PhD Admissions Committee (August 2025 - August 2026)

Member of University of Waterloo CS Awards Committee (August 2023 - Present)

Member of University of Waterloo CS Graduate Committee (August 2024 - August 2025)

Member of University of Waterloo CS Women in Computer Science Committee (August 2023 - August 2024)

Member of University of Waterloo CS School Advisory Committee on Appointments (August 2022 - August 2023)

Member of University of Waterloo CS Equity, Diversity, and Inclusion Committee (August 2020 - August 2022)

Member of University of Waterloo CS Graduate Recruitment Committee (August 2019 - August 2020)

Vector Institute Faculty Member Committee (2026)

Vector Institute Faculty Affiliate Committee (2025)

Reviewer for Vector Institute Visiting Researcher Program (Winter 2024, Spring 2024)

Reviewer for Vector Scholarship in AI (Spring 2023)

Organizer for CIFAR Deep Learning + Reinforcement Learning Summer School (Fall 2023 - Summer 2024)

THESIS COMMITTEES

PhD Thesis: Amit Levi (Waterloo), Vikrant Singhal (Northeastern University), Guojun Zhang (Waterloo), Jimit Majmudar (Waterloo), Tosca Lechner (Waterloo), Nathan Harms (Waterloo), Kelly Ramsay (Waterloo), Bailey Kacsmar (Waterloo), Tim Dockhorn (Waterloo), Yiwei Lu (Waterloo), Aseem Baranwal (Waterloo), Ruicheng Xian (UIUC), Saber Malekmohammadi (Waterloo), Shubhankar Mohapatra (Waterloo), Youssef Allouah (EPFL)

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